ALEJANDRO VILLARRUBIA

Technical Sales Engineer · Software Engineer

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PROFESSIONAL PROFILE

Computer Engineer with experience in software development, hardware/software integration, and coordination across engineering, sales, and customers. Currently a Technical Sales Engineer at ALTEN Spain for MAHLE, specialising in e-bike propulsion systems. Previous backend experience with Java, MongoDB, and SQL. Particularly interested in building systems from scratch, artificial intelligence, algorithms, and personal technical projects.

PROFESSIONAL EXPERIENCE

Technical Sales Engineer

Jul. 2026 – Present

ALTEN Spain · Client: MAHLE | Palencia, Spain

- Coordinate the planning and monitoring of e-bike propulsion system fleets with Program and Product Management, managing resources, deliverables, and technical documentation.
- Validate the technical compatibility of customer bicycle frames (Customer Frame Analysis) with MAHLE systems before production.
- Participate in test rides and motor map configuration (power and torque), working with Battery and Control Systems to evaluate motors and batteries.
- Act as the link between sales, engineering, and customers, preparing technical reports and managing samples and associated documentation.

Software Engineer

Oct. 2025 – Jul. 2026

NTT DATA · Spain (national distributed team)

- Independently developed a new product feature from scratch based on an existing solution, participating in the full delivery cycle in an Agile/Scrum environment.
- Refactored code, fixed defects, and resolved production incidents to improve maintainability and reduce technical debt within a team of more than 20 engineers.
- Participated in solution design, implementation, testing, debugging, code review, and deployment.

Java · SQL · MongoDB · Git · Agile/Scrum

Software Development Intern

Apr. 2025 – Jul. 2025

REDALSA, S.A.

- Designed and implemented a LiDAR-based railway object detection system from scratch, including distance measurement, mapping, and real-time collision and trajectory calculations.
- Independently integrated hardware and software, from signal acquisition and processing to data visualisation and interpretation.

Python · C · LiDAR · Embedded Systems · Data Acquisition

Active Committee Member

Apr. 2024 – Oct. 2025

RITSI · Spanish national association of Computer Engineering students

- Participated in four national committees involving more than 30 universities, including the organisation of national conferences and maintenance of technology tools.

EDUCATION

B.Sc. in Computer Engineering

2021 – 2026

Universidad Europea Miguel de Cervantes (UEMC)

Master's Program in Artificial Intelligence and Innovation

2025

Microsoft · Founderz | Postgraduate training

Science Baccalaureate

2015 – 2021

IES Jorge Manrique

SELECTED PROJECTS

ChessBot – Chess Engine 2026 – Present

Personal project · C++20 · UCI · Local AI

- Custom engine for playing and analysing positions, with search and evaluation systems and UCI communication with interfaces and other engines.
- Training infrastructure using self-play, matches against other engines, and experiments with positional evaluation and NNUE networks.
- Windows application and optional local LLM integration to explain engine calculations. Training and evaluation are ongoing; no certified Elo rating.

Pipα – Local Assistant and Hardware Controller 2026 – Present

Personal project · Python · ESP32-S3 · Windows · Swift

- Local agent for automating applications, audio, system status, and control actions from an external device, with confirmable commands.
- Hardware/software integration: touchscreen, local voice, encrypted USB communication, and Wake-on-LAN. Experimental iOS client.

Email Phishing Detection System 2026

Bachelor's thesis · Python · scikit-learn · Streamlit

- Heuristic analysis of 28 signals (headers, URLs, HTML, and SPF/DKIM/DMARC) and an MLP neural classifier trained on real-world datasets.
- TF-IDF + MLP pipeline, web interfaces, Gmail extension, tests, and model persistence; analysis in Spanish and English.

AQUILAE – Roman Empire Risk Simulator 2025

Master's thesis collaboration · Java · Swing · Maven

- Designed and implemented the complete simulator with a historical vector map of 28 provinces and a Java Swing interface, supporting a master's thesis at the University of Valladolid.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, C#, JavaScript, HTML, PHP, SQL.

Data and tools: SQL, MongoDB, Git, Agile/Scrum, Office 365.

Specialisms: Search algorithms, UCI, AI/ML, hardware/software integration, LiDAR, embedded systems.

CERTIFICATIONS AND LANGUAGES

B2 First – Cambridge University · Digital Skills Certificate (tuCertiCyL) · Regional Chess Coach · Regional Chess Arbiter.

Spanish: Native · English: B2 (Cambridge First) · German: Basic.

LEADERSHIP AND COMMUNITY

University Student Representative · School, Degree and Class Oct. 2021 – Oct. 2025

UEMC · Institutional representation and liaison with academic leadership and the dean's office.

UEMC Representative at RITSI Oct. 2023 – Nov. 2025

Official delegate to Spain's national Computer Engineering student association; committees and conferences.

Secretary · Club Palentino de Ajedrez Mar. 2026 – Present

Management and organisation of activities for a club with over 40 years of history.

Founder and Coordinator Jun. 2024 – Present

UEMC Book Club · UEMC Chess School.